NPDE Note

1 Definitions, Classification of PDEs

1.1 Big O, Classification

Big $\mathcal{O}$: $f(t)=\mathcal{O}(g(t))$ as $t\to t_0$ if $\exists\varepsilon>0, C>0$ s.t. $\forall|t-t_0|<\varepsilon, |f(t)|\le C|g(t)|$

$\Leftrightarrow \lim_{t\to t_0}[f(t)/g(t)]$ bdd   ps: If not say $t\to t_0$, assume $t\to 0$

More on Big $\mathcal{O}$: a. $F(x)=G(x)+\mathcal{O}(H(x))$ if $F(x)-G(x)=\mathcal{O}(H(x))$

b. $F(x)=\mathcal{O}(G(x))+\mathcal{O}(H(x))$ if $F(x)=\mathcal{O}(\max\{|G(x)|,|H(x)|\})$

c. $F(x,y)=\mathcal{O}(x^m)+\mathcal{O}(y^n)$ if [Given $G(x)=\mathcal{O}(x^{\frac{m}{n}})\implies F(x,G(x))=\mathcal{O}(x^m)$], 且另一项同理

Classification: For $au_{xx}+bu_{xy}+cu_{yy}+du_x+eu_y+fu+g=0$ $(a,b,c)\neq(0,0,0)$

1. **Hyperbolic**: $b^2-4ac>0$ Admits non-trivial plane wave solutions
2. **Parabolic**: $b^2-4ac=0$ (对变量 x,y 进行缩放后不会改变 PDE 类型)
3. **Elliptic**: $b^2-4ac<0$ No non-trivial plane wave solutions

1.2 Norm

Weighted 2-Norm: $\|\mathbf{u}\|_{2,\Delta x}=\sqrt{(\Delta x)^d}\|u\|_2$ where d s.t. $\mathbf{u}:\mathbb{R}^d\to\mathbb{R}^n$

- For $d=1$ and $\Delta x=\frac{D}{N}$, then: $\forall i, u_i=\mathcal{O}((\Delta x)^p)\Rightarrow\|\mathbf{u}\|_{2,\Delta x}=\mathcal{O}((\Delta x)^p)$
- For A symmetric: $\|A^{-1}\mathbf{w}\|_{2,\Delta x}\le\frac{1}{|\lambda_1|}\|\mathbf{w}\|_{2,\Delta x}$ $0<|\lambda_1|\le\cdots\le|\lambda_n|$

Property: If $\|\mathbf{u}\|_2\le\|\mathbf{w}\|_2$, then $\|\mathbf{u}\|_{2,\Delta x}\le\|\mathbf{w}\|_{2,\Delta x}$

New Norm: The norm $\|\mathbf{u}\|:=\max_n\|\mathbf{u}^n\|_{2,\Delta x}$ where $\mathbf{u}^n=[u_1^n,u_2^n,\ldots,u_{M-1}^n]^\top$

Which means: Infinity norm in time, weighted 2-norm in space.   主要用于双变量 PDE

- For $\tau_m^n=\mathcal{O}((\Delta x)^a)+\mathcal{O}((\Delta t)^b)$, $\Delta x=\frac{b-a}{M}$, $\Delta t=\frac{T}{N}$, assume $\Delta t=K(\Delta x)^\beta$, then: $\max_n\|\underline{\tau}^n\|_{2,\Delta x}=\mathcal{O}((\Delta x)^{\min\{a,b\beta\}})$

1.3 Linear Algebra

Definite: Positive Definite: $\lambda_i>0,\forall i$ Negative Definite: $\lambda_i<0,\forall i$

Symmetric: If A symmetric, then $A=Q\Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.

Normal: Real matrix A is normal if $AA^\top=A^\top A$ The following are properties of it:

1. A is orthogonally diagonalizable, i.e. $A=Q\Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.
2. Exists an eigenvectors of A: $\{\mathbf{v}_i\}$ s.t. $\mathbf{v}_i^*\mathbf{v}_j=\delta_{ij}$ and Symmetric $\Rightarrow$ Normal

2 Global/Local Error, Convergence, Consistency, Stability

Exact & Approx. Sol: U_n: approx. n-th time-step sol; u_n: exact sol at n-th time-step.

Global Error $(e_n,\mathbf{e}(\Delta t))$: $e_n=U_n-u(t=n\Delta t)$ $\mathbf{e}(\Delta t)=[e_0,e_1,\ldots,e_N]^\top$

Convergence: A numerical method is convergent if $\lim_{\Delta t\to 0}\|\mathbf{e}(\Delta t)\|=0$

If $\lim_{\Delta t\to 0}\frac{\|\mathbf{e}(\Delta t)\|}{(\Delta t)^p}=C$ (i.e. $\lim_{\Delta t\to 0}\|\mathbf{e}(\Delta t)\|=\mathcal{O}((\Delta t)^p)$) $\Rightarrow$ p-th order accurate

Local Truncation Error (τ_n): 数值分析的离散迭代公式, 将 U_n 替换为 u_n 后的误差 *(LHS - RHS)*.

- e.g. For $\frac{U_{n+1}-U_n}{\Delta t}=f(U_n)$, $\tau_{n+1}=\frac{u_{n+1}-u_n}{\Delta t}-f(u_n)$ $\tau_0=0$
- e.g. 我们可以把线性数值方法写成 $Au=b$, A matrix, $u=[U_1,\ldots,U_{N-1}]^\top$.

  Then: $\tau=Au_{\text{exact}}-b$ where $u_{\text{exact}}=[u_1,\ldots,u_{N-1}]^\top$, $\tau=[\tau_1,\ldots,\tau_{N-1}]^\top$

ps: $Ae=-\tau$ **where** $e=u-u_{\text{exact}}$ Define $\tau(\Delta t)=[\tau_1,\ldots,\tau_{N-1}]^\top$

Consistency: If $\lim_{\Delta t\to 0}\|\tau(\Delta t)\|=0$, the method is consistent.

Linear Equation: Goal: Finally change ODE/PDE into $Au=b$ where A **is matrix**, $u=[U_1,\ldots,U_{N-1}]^\top$

Stability: A numerical method is stable if $\exists\Delta x_0$ s.t. For the Linear System $Au=b$:

1. $A(\Delta x)$ is invertible $\forall\Delta x<\Delta x_0$
2. $\|A^{-1}(\Delta x)\mathbf{w}\|\le C\|\mathbf{w}\|$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$

Property: Consistent + Stable $\Rightarrow$ Convergent (same norm!)   and $\|\mathbf{e}(\Delta t)\|=\mathcal{O}(\|\tau(\Delta t)\|)$

3 Intro & Method of Lines

Taylor Series: $f(t+\Delta t)=f(t)+f'(t)\Delta t+\frac{1}{2}f''(t)(\Delta t)^2+\cdots+\mathcal{O}((\Delta t)^N)$ **ps:** $u_{n+1}=u(n\Delta t+\Delta t)$

$$f(t)=f(a)+f'(a)(t-a)+\frac{f''(a)}{2!}(t-a)^2+\cdots+\frac{f^{(N-1)}(a)}{(N-1)!}(t-a)^N+\mathcal{O}((t-a)^N)$$

Euler Methods: Forward: $\frac{du}{dt}|_{n\Delta t}\approx\frac{U_{n+1}-U_n}{\Delta t}$; Backward: $\frac{du}{dt}|_{n\Delta t}\approx\frac{U_n-U_{n-1}}{\Delta t}$

Leapfrog: $\frac{du}{dt}|_{n\Delta t}\approx\frac{U_{n+1}-U_{n-1}}{2\Delta t}$ **Trapezoidal:** $\frac{1}{2}[\frac{du}{dt}|_{n\Delta t}+\frac{du}{dt}|_{(n+1)\Delta t}]\approx\frac{U_{n+1}-U_n}{\Delta t}$

Centred Difference: $\frac{d^2u}{dt^2}|_{n\Delta t}\approx\frac{U_{n-1}-2U_n+U_{n+1}}{(\Delta t)^2}$

Method of Lines: Using $U_m(t)\approx u(a+m\Delta x,t)$, $U_m^n\approx u(a+m\Delta x,n\Delta t)$ a 为空间起点, 不重要

1. **For 1D: a.** 先对空间 x 离散化, 使用 U_m 表达,   此时 $\frac{d\mathbf{U}}{dt}=\ldots$ where $\mathbf{U}=[U_1(t),U_2(t),\ldots,U_{M-1}(t)]^\top$

   b. 对时间 t 使用 ODE 离散化, 使用 U_m^n 表达, 得到 Linear Equation.
2. Example: Crank-Nicolson for Heat Eq. $u_t=\kappa u_{xx}$: 对 x 用 centred difference; 对 t 用 trapezoidal:

   $$\frac{U_m^{n+1}-U_m^n}{\Delta t}=\frac{\kappa}{2}\left(\frac{U_{m+1}^{n+1}-2U_m^{n+1}+U_{m-1}^{n+1}}{(\Delta x)^2}+\frac{U_{m+1}^n-2U_m^n+U_{m-1}^n}{(\Delta x)^2}\right)$$

   $$\Rightarrow -rU_{m-1}^{n+1}+(2+2r)U_m^{n+1}-rU_{m+1}^{n+1}=rU_{m-1}^n+(2-2r)U_m^n+rU_{m+1}^n,\quad r=\frac{\kappa\Delta t}{(\Delta x)^2}$$

Solve it: By iteration: e.g. $A\mathbf{U}^{n+1}=\mathbf{B}\mathbf{U}^n$,   where $\mathbf{U}^n=[U_1^n,U_2^n,\ldots,U_{M-1}^n]^\top$.

Truncation for MoL: 对于 τ_m, 离散 x 后 U_m 替换为 u_m; 对于 τ_m^n, 再离散 t 后 U_m^n 替换为 u_m^n.

e.g. For Crank-Nicolson above: $\tau_m=\frac{\partial u_m}{\partial t}-\kappa\frac{u_{m+1}-2u_m+u_{m-1}}{(\Delta x)^2}$

e.g. For Crank-Nicolson above: $\tau_m^n=\frac{u_m^{n+1}-u_m^n}{\Delta t}-\frac{\kappa}{2}(\cdots)$

4 Lax-Richtmyer Stability, von Neumann Analysis

Abs. Stability for ODEs: For ODEs $\frac{d\mathbf{u}}{dt}=A\mathbf{u}$, $\mathbf{u}$ vector.   A 1-step method for t (i.e. $\mathbf{U}^{n+1}=B\mathbf{U}^n$)

It's absolutely stable if $\|B\mathbf{w}\|_2\le\|\mathbf{w}\|_2$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$

Property: If B normal, then absolutely stable $\Leftrightarrow|\lambda_i(B)|\le 1,\forall i$

Property: Absolute stability $\Rightarrow$ Lax-Richtmyer stability in 2-norm and weighted 2-norm

Lax-Richtmyer Stability: For PDE which discrete form $\frac{1}{\Delta t}\mathbf{U}^{n+1}=\frac{1}{\Delta t}B\mathbf{U}^n+\mathbf{F}^n$, B fixed, $\mathbf{F}$ can change.

It's Lax-Richtmyer stable if $\|B^n\mathbf{w}\|\le C\|\mathbf{w}\|$ for all $\mathbf{w}\in\mathbb{R}^{N-1}$, $n\le\frac{T}{\Delta t}=N$

Property: If consistent with $\max_n\|\underline{\tau}^n\|_{2,\Delta x}=\mathcal{O}((\Delta x)^\beta)$ and L-R stable $\|B^n\mathbf{w}\|_{2,\Delta x}\le C\|\mathbf{w}\|_{2,\Delta x}$, then, it convergents with $\max_n\|\mathbf{e}^n\|_{2,\Delta x}=\mathcal{O}((\Delta x)^\beta)$ Remark: **Always Assume $\Delta t=K(\Delta x)^\alpha$**

von Neumann Analysis: PDE: **1.** *periodic* BCs in spatial x, length D. **2.** 满足 L-R 中 PDE 框架, **3.** $\mathbf{F}^n=0$

$\Rightarrow$ Let $v_{k,m}=\exp(2\pi ik\frac{m\Delta x}{D})$ for $m=0,1,\ldots,M-1$.   Then vector $\mathbf{v}_k$ 为 B 的正交特征向量.

$\star$ By $U_m^n=\exp(2\pi ik\frac{m\Delta x}{D})$, $U_m^{n+1}=\alpha_kU_m^n$, 可得特征值(放大因子) α_k 表达式 (i.e. $B\mathbf{v}_k=\alpha_k\mathbf{v}_k$).

$\Rightarrow$ If all α_k for $k=0,1,\ldots,M-1$ can be found $\Rightarrow B$ is normal and then:

If $|\alpha_k|\le 1,\forall k\Rightarrow B$ is absolutely stable and L-R stable in weighted 2-norm

Remark: 假设 $\Delta t=K(\Delta x)^\alpha$ 会得 K,α 的限制条件, || For $\mathbf{F}^n\neq 0$, 讨论 $\mathbf{F}^n=0$ 足够.

5 Boundary Condition

2nd ODE: For $u_xx=\beta(x)$ with $x\in(a,b)$, there are three types of BCs:

1. **Two Dirichlet**: $u(x=a)=c$ $u(x=b)=d$
2. **One Dirichlet, One Neumann**: $u(x=a)=c$ $u_x(x=b)=d$ or $u_x(x=a)=c$ $u(x=b)=d$
3. **One Neumann, One Constraint**: $u_x(x=a)=c$ $\int_a^b u(x)dx=d$

Heat PDE: For $u_t=\kappa u_{xx}$ with $x\in(a,b)$, there are three types of BCs: $\star$ **Basic IC**: $u(x,t_0)=P(x)$

1. **Two Dirichlet**: $u(x=a,t)=c$ $u(x=b,t)=d$
2. **One Dirichlet, One Neumann**: $u(x=a,t)=c$ $u_x(x=b,t)=d$ or $u_x(x=a,t)=c$ $u(x=b,t)=d$
3. **Two Neumann**: $u_x(x=a,t)=c$ $u_x(x=b,t)=d$

6 Appendix

x,x^2,x^3 **as** $x\to 0$, x^{-1},x^{-2},x^{-3} **as** $x\to$ **inf**

Types of PDEs: Laplace Eq.: $u_{xx}+u_{yy}=0$ (E)   Poisson Eq.: $u_{xx}+u_{yy}=f(x,y)$ (E)

Helmholtz Eq.: $u_{xx}+u_{yy}+\alpha u=0$, $\alpha\neq 0$ (E)   Heat Eq.: $u_t=\kappa(u_{xx}+u_{yy})$, $\kappa>0$ (P)

Wave Eq.: $u_{xx}-u_{yy}=f(x,y)$ (H)   Advection Eq.: $u_t+vy_x+wu_y=0$ (H)

Taylor Series: $f(t+\Delta t)=f(t)+f'(t)\Delta t+\frac{1}{2}f''(t)(\Delta t)^2+\cdots+\mathcal{O}((\Delta t)^N)$ **ps:** $u_{n+1}=u(n\Delta t+\Delta t)$

$u_m^{n+1}=u_m^n+\Delta t(u_t)_m^n+\frac{(\Delta t)^2}{2}(u_{tt})_m^n+\cdots$ where $(u_t)_m^n=\frac{\partial u}{\partial t}(a+m\Delta x,n\Delta t)$

$u_{m+1}^n=u_m^n+\Delta x(u_x)_m^n+\frac{(\Delta x)^2}{2}(u_{xx})_m^n+\cdots$ where $(u_x)_m^n=\frac{\partial u}{\partial x}(a+m\Delta x,n\Delta t)$

Useful Tools: If $e^n=Ae^{n-1}+B\tau^n$, then $e^n=A^ne^0+\sum_{k=1}^n A^{n-k}B\tau^k$, A: amplification factor.

Others: Periodic boundary conditions for x: $u(a,t)=u(b,t)$, $u_x(a,t)=u_x(b,t)$ a,b 边界.

Heat Equation: For $u_t=\kappa u_{xx}$

	Forard Euler	Backward Euler	Crank-Nicolson
Solve Matrix?	No	Yes	Yes
r_m^n	$\mathcal{O}(\Delta t)+\mathcal{O}((\Delta x)^2)$	$\mathcal{O}(\Delta t)+\mathcal{O}((\Delta x)^2)$	$\mathcal{O}((\Delta t)^2)+\mathcal{O}((\Delta x)^2)$
Stability Condition	$\kappa\frac{\Delta t}{(\Delta x)^2}\le\frac{1}{2}$	None	None